



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 • STANDARD 6.GR.2

# Volume of Rectangular Prisms

Lesson 10-2 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 3 Enrichment



### TODAY'S OBJECTIVES

**Content Objective:** I can find the volume of a rectangular prism, including ones with fractional edge lengths, using  $\text{base area} \times \text{height}$ .

**Language Objective:** I can explain my work using the words volume, rectangular prism, dimensions, and base area.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Volume

Rectangular prism

Cubic units

Dimensions

Net

Base area



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The amount of space inside a three-dimensional solid is its

2

A solid with six rectangular faces is a .

3

The unit used to measure volume is .

4

The measurements of length, width, and height of a solid are its

5

A flat pattern that folds into a solid is a .

- 6 The area of the bottom face of a prism, multiplied by height to find volume, is the

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

COMPARE

**Capsule A ( $2 \times 1.5 \times 1$ ) and Capsule B ( $3 \times 1 \times 1$ ) both have a volume of  $3 \text{ ft}^3$ .  
How did your multiplication produce the same volume from different dimensions?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **Volume** — How much space is inside a solid shape.  
*Volumen — Cuánto espacio hay dentro de una figura sólida.*

☐ **Rectangular prism** — A solid box shape with six flat rectangle sides.  
*Prisma rectangular — Una figura sólida en forma de caja con seis lados rectangulares.*

☐ **Cubic units** — The units used to measure space inside, like cubic inches.  
*Unidades cúbicas — Las unidades para medir el espacio interior, como pulgadas cúbicas.*

☐ **Dimensions** — How long, how wide, and how tall a shape is.  
*Dimensiones — Qué tan largo, ancho y alto es una figura.*

☐ **Base area** — The area of the bottom of a solid.  $\text{Volume} = \text{base area} \times \text{height}$ .  
*Área de la base — El área del fondo de un sólido.  $\text{Volumen} = \text{área de la base} \times \text{altura}$ .*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**Both capsules have  $V = \underline{\hspace{1cm}} \text{ ft}^3$  because  $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$ .**

*Ambas cápsulas tienen  $V = \underline{\hspace{1cm}} \text{ ft}^3$  porque  $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$ .*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** You still multiply length  $\times$  width  $\times$  height even when an edge is a fraction or decimal.

**Evidence:** Students explain the formula stays the same ( $V = l \times w \times h = 2 \times 1.5 \times 1 = 3 \text{ ft}^3$ ) and that a fractional edge just means multiplying with a fraction or decimal.

**Reasoning:** This evidence connects the definition of volume to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **Both capsules have  $V = \underline{\hspace{1cm}} \text{ ft}^3$  because  $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$ .**

*Ambas cápsulas tienen  $V = \underline{\hspace{1cm}} \text{ ft}^3$  porque  $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$ .*

- **The dimensions are different but the  $\underline{\hspace{1cm}}$  is the same.**

*Las dimensiones son diferentes pero el  $\underline{\hspace{1cm}}$  es el mismo.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is  $\underline{\hspace{1cm}}$ .**

*Mi afirmación es  $\underline{\hspace{1cm}}$ .*

- **I know because  $\underline{\hspace{1cm}}$ .**

*Lo sé porque  $\underline{\hspace{1cm}}$ .*

- **So  $\underline{\hspace{1cm}}$ .**

*Entonces  $\underline{\hspace{1cm}}$ .*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.



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## Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Dimensions
5. **Notes 5:** Net
6. **Notes 6:** Base area
7. **Watch for this mistake:** *A common mistake in Volume of Rectangular Prisms is rounding a fractional edge length to the nearest whole number before multiplying, instead of using the exact fraction or decimal. For a box that is  $2\text{ ft} \times 1.5\text{ ft} \times 1\text{ ft}$ , a student might round 1.5 down to 1 and compute  $2 \times 1 \times 1 = 2\text{ ft}^3$ , but the correct volume uses the exact edge length:  $2 \times 1.5 \times 1 = 3\text{ ft}^3$  — a full cubic foot larger. Before you submit, ask: 'Did I multiply using the exact fractional or decimal edge length, or did I round it off first?'*
8. **Try It 1:** A. Box A, because 50 cubic in is more than 48 cubic in — Box A:  $5 \times 5 \times 2 = 50\text{ cubic in}$ . Box B:  $4 \times 4 \times 3 = 48\text{ cubic in}$ .  $50 > 48$ , so Box A holds more.
9. **Try It 2:** A. 18 cubic in —  $V = 4 \times 3 \times 1.5 = 12 \times 1.5 = 18\text{ cubic in}$ . A fractional edge does not change the steps.